



ME 615: Applied Computational Fluid Dynamics

Course Code	ME 615
Course Name	Applied Computational Fluid Dynamics
L-T-P-C	2.5-0.5-0-3
Intended for	BTech 4th Year/MS/PhD
Distribution	Elective
Approval	2nd Adhoc Senate (May 2012)

Course Contents

- **Introduction**
Basics of heat transfer, fluid flow; Mathematical description of fluid flow and heat 5 transfer: conservation equations for mass, momentum, energy and chemical species, classification of partial differential equations, coordinate systems. [Lectures]
- **Discretization techniques**
Discretisation techniques using finite difference methods: Taylor-Series and control volume formulations; Finite element discretization techniques. [Lectures]
- **Modelling of diffusion problems using finite volume method**
One dimensional steady state diffusion problems; discretization technique; Solution methodology for linear and non-linear problems: Point-by-point iteration, TDMA; Two and three dimensional discretization; Discretization of unsteady diffusion problems: Explicit, Implicit and Crank-Nicolson's algorithm; stability of solutions. [Lectures]
- **Modelling of Convection- Diffusion Problems**
One dimensional convection-diffusion problem: Central difference scheme; Discretization based on analytical approach (exponential scheme); Hybrid and power law discretization techniques; Higher order schemes (QUICK algorithm). [Lectures]
- **Flow modeling**
Discretization of incompressible flow equations; Pressure based algorithm: SIMPLE, SIMPLER etc; Unstructured grids; Introduction to FVM with unstructured grids; Introduction to turbulence modeling; Large Eddy Simulation (LES); Direct Numerical Simulation (DNS). [Lectures]
- **Projects / Exercises/ Publications**
Solving simplified problems: formulation, discretization with coarse grids, applying appropriate boundary and initial conditions and solving by hand calculations; Solving practical problems through software: writing user sub-routines; postprocessing and interpretation of results. [Lectures]

Text and Reference Books

1. S. V. Patankar, **Numerical Heat Transfer and Fluid Flow**, Hemi sphere Publishing Corporation, 1980.
2. D. A. Anderson, J. C. Tannehill , and R. H. Pletcher, **Computational Fluid mechanics and Heat Transfer**, Hemi sphere Publishing Corporation, 1984.
3. J. H. Ferziger and M. Peric, **Computanional Methods for Fluid Dynamics**, 2nd Edition, Springer, Berlin, 1999.
4. H. K. Versteeg and W. Malalasekera, **An Introduction to Computational Fluid Dynamics: The Finite Volume Method**, Pearson.